

Chromatic Contention

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In Chromatopia, there are two species of chromatopes, *chromatopods* and *chromataceans*, each of which are one of n possible colors. As an annual ritual, the Council of Colors arranges chromatopes in a line. However, because chromatopods and chromataceans are sworn enemies, chromatopes of the same color but different species are hostile towards each other and refuse to stand next to one another. In particular, if a chromatopod is next to at least one identically-colored chromatacean, the chromatopod and one of the adjacent chromataceans both leave the line, and vice versa. As per the sacred ritual, the Council has arranged them according to the following rules:

1. There are no chromatopods adjacent to identically-colored chromataceans.
2. If all chromatopes of a color are removed from the line, then all the other creatures will eventually leave the line, as the absence of other creatures causes them to be adjacent to a hostile creature.

For example, when $n = 2$, a valid line is $a_1 a_2 b_1 b_2$, since removing either a_1 and b_1 or a_2 and b_2 will result in the remaining two being of the same color but different species, causing them to leave the line, resulting in an empty line.

What does such a line look like for $n = 3$? What about $n = 4$? How can this be generalized to arbitrary values of n ?